

PATRICK WOODBURN

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EDUCATION

Harrisburg University Of Science And Technology

Bachelor Computer Information Sciences

2010-09-01 - 2014-05-01

EXPERIENCE

Samsung Smartthings

Senior Software Engineer

2022-08-01 - 2025-03-31

- Created and maintained microservices that run on AWS to support the SmartThings platform.
- Used Terraform to manage infrastructure as code for the SmartThings platform.
- Optimized performance of jvm and ec2 instances to reduce costs and increase reliability of the SmartThings platform.
- Leveraged agentic workflows to accelerate api migration to newer versions of Java and more modern frameworks.
- Maintained api documentation for services to ensure that they are up to date and accurate for both internal and external use.
- Design Dynamo Tables for optimal performance and cost efficiency as well as ease of use for developers.
- Enhanced monitoring and alerting systems to ensure that issues with the SmartThings platform are quickly identified and resolved.
- Handled on call responsibilities to maintain the reliability and uptime of the SmartThings platform.

Fort Robotics

Senior Software Engineer

2021-06-06 - 2022-07-31

- Worked with yocto to create embedded linux distributions for custom hardware.
- Created standard build environments to enable developers to build consistent software.
- Increased performance of CMake build tools.
- Created chip flashing tools for low level radio interfaces.
- Worked on embedded communications system using zmq and coap.
- Set up automatic build systems for multiple code repositories in multiple programming languages.
- Improved code stability through the implementation of static code analysis.
- Transitioned dependency management to more manageable system allowing for faster more reliable software builds.
- Worked on embedded hardware to increase safety in industrial environments.

Thomas Jefferson University - Dice Group

Software Engineer

2017-12-21 - 2021-06-04

- Worked on software that drove the first ever HIPPA compliant smart speaker.
- Built APIs that allowed for internet connected devices to communicate between each other for the purpose of controlling equipment in a hospital setting.
- Wrote Linux scripts used to manage a suite of software focused on the embedded platform.
- Built a system for intercommunication between APIs and firewall protected embedded systems.
- Designed a system for delivering software updates leveraging MQTT events.
- Used Google Cloud Platform to control and maintain numerous Jefferson IoT projects.
- Improved data flow between existing APIs to ensure negligible delay between interconnected systems.
- Lead conversations on uses, pros and cons, and implementation of IoT networks including LoRaWAN, Bluetooth, WiFi, and cellular within the engineering team.
- Designed and created prototypes of simple electrical circuits for small batch products on experimental IoT and embedded system projects.
- Designed 3D printed short run rapid prototype housings for early phase projects during the proof of concept stage.

- Created an interface for controlling Docker over the MQTT network.
- Maintained Docker images for testing embedded software in order to eliminate the need for physical hardware.
- Created workflows for using Docker on embedded systems.
- Created systems to automatically and dynamically deploy software to embedded hardware.
- Implement Redis as a low lag reliable cross application key store and communication bus solution.

Comcast

Software Engineer

2015-03-01 - 2017-05-19

- Operated within a team of developers to develop and test a new home automation system.
- Worked with internet connected devices to automate and secure user's homes.
- Developed android apps that could run against a users unique home automation system.
- Used testing tools such as junit and espresso to create test code to run against android apps for continuous integration.
- Built test kits to simulate devices in a typical user's home.
- Created technical documentation for both installation technicians and tech support to use when assisting users.
- Set up continuous integration environment using Jenkins.
- Scripted tools to deploy to and debug on custom android firmware.
- Developed software to communicate over wireless interfaces such as bluetooth, WIFI, and Zigbee.

Franklin Institute

Science Educator

2006-04-01 - 2019-06-30

- Planned lessons on the topics of electrical and software development.
- Taught students how to program and build circuitry required for multiple applications of robotics.
- Worked with students to build robots for various competition and museum displays.
- Documented use and maintenance of students' projects to be handed off to groups tasked with operation through museum.
- Built software tools that the students can use to better debug their projects.
- Presented and demonstrated robotics projects to large groups of people.
- Operated robots in and around the museum.

Independence Blue Cross

Software Engineer

2014-08-01 - 2015-02-01

- Implemented given designs so that users may have the most comfortable experience possible when accessing their insurance information.
- Created cross-platform mobile apps that could be rebranded for multiple insurance companies.
- Ported functionality of apps between incompatible code repositories.

BioHitech America

Intern

2014-01-01 - 2014-07-01

- Developed software to run on headless industrial machinery and perform analytics regarding their use.
- Set up and maintained Linux devices to be used in embedded industrial applications.
- Created static environment configuration for linux machinery and virtual machines.
- Developed Kiosk software to be able to track metrics from items scanned with nfc chips.
- Developed webpages to pull from various RESTful APIs.
- Created firmware to drive touchscreen pannels.
- Researched and retrofitted new hardware for industrial-grade machinery.

SKILLS

Software Engineering:	Java, spring, zig, C, C++, CMake, Python, Android, HTML, Object Oriented Programming, Java
Electrical Design:	Circuit Design, Prototyping, soldering, CAD, PCB design, 3D printing